Data_ip EDA Report

I have looked at the data in data_ip table and tried to do some exploratory data analysis, basically just print and plot all the possible things that I could think of to possible get some insight from the data.

The data is organized as follows: it has four columns station_id, year, ice_on, ice_off

The station_id is a unique identifier of a measuring station, it is almost equivalent to lake id that is almost all lakes have only one stations, there is totally 1197 lakes and this is the distribution, how many stations they have:

stations	lakes
1	1150
2	31
3	9
4	3
5	2
6	1
12	1

And here are the particular lake id's that have two and more stations.

Lakes with multiple stations:
2 stations: 1009, 1048, 159, 166, 209, 226, 240, 267, 284, 298, 3, 315, 360, 374, 39, 390, 425, 550, 623, 630, 631, 633, 636, 650, 663, 675, 699, 706, 720, 896, 938
3 stations: 15, 158, 173, 176, 259, 48, 522, 549, 930
4 stations: 516, 659, 977
5 stations: 450, 590
6 stations: 571
12 stations: 296

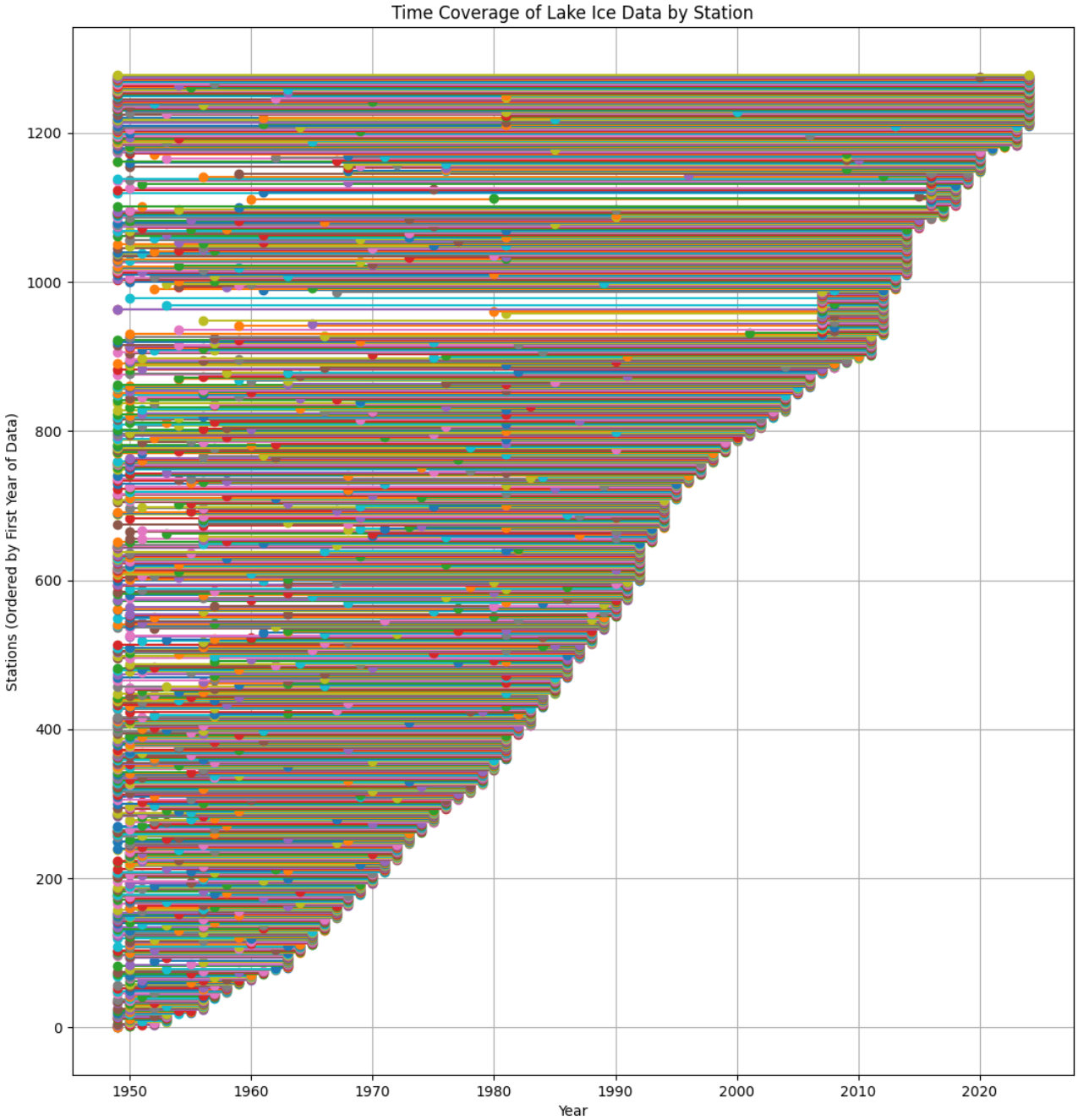
We see that the majority of lakes 1150 have only one station, a few 31 have two, then it decreases rapidly. There is even one lake with 12 stations.

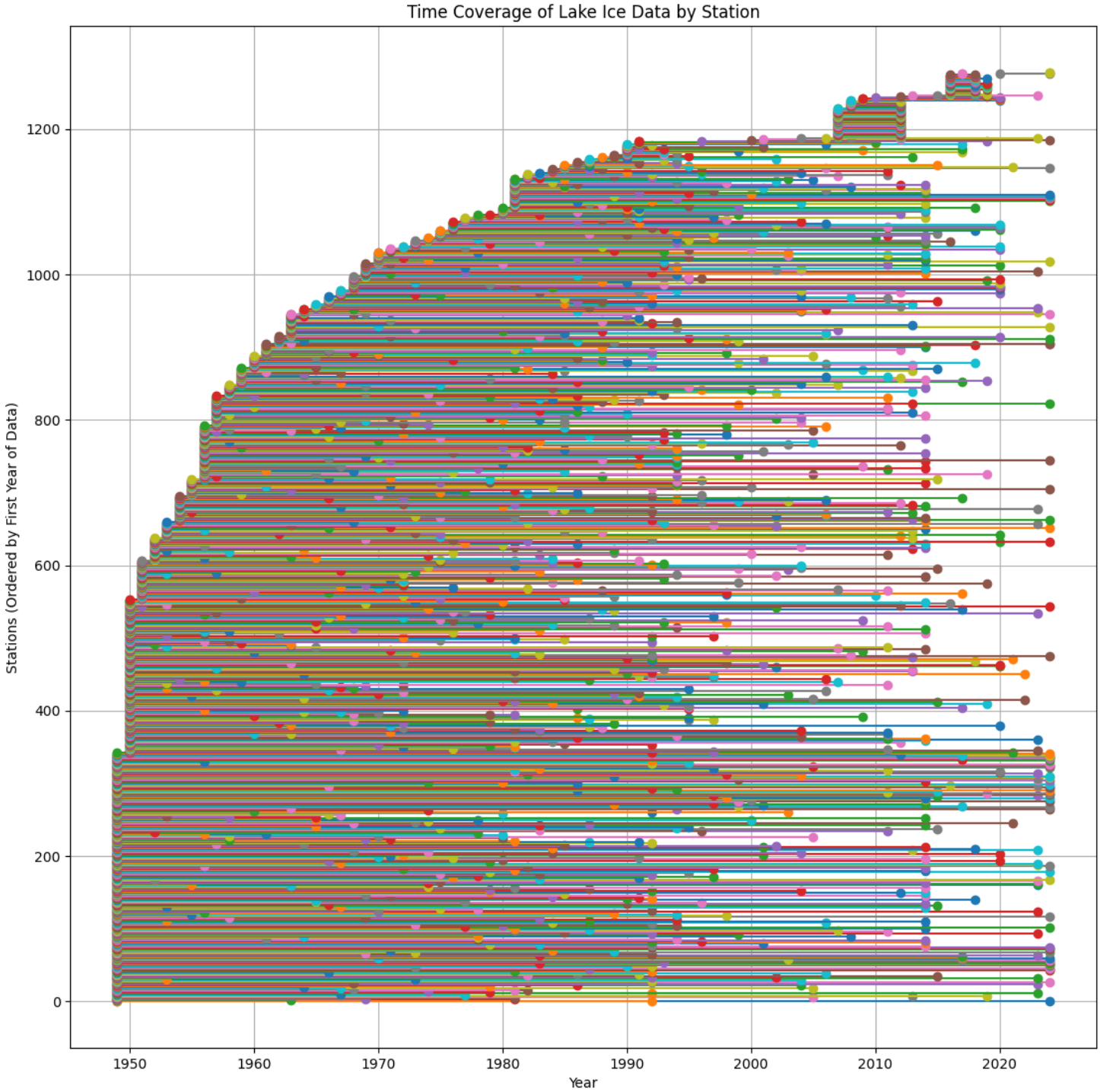
Then we have column year, which is the year of the measurement, and then we have two columns: ice_on and ice_off. Ice on is the day of the previous year when the lake froze and ice off the day when the ice melted. There are no missing data in the column year, that's good. That would be bad to have some measurement without knowing from which year they are, but there are some missing values for the ice_on and ice_off values, Here is the amount of missing data:

Total records: 37324
Records with both ice_on and ice_off: 30877
Records with missing ice_on: 4834 (12.95%)
Records with missing ice_off: 1613 (4.32%)
Records with both ice_on and ice_off missing: 0 (0.00%)

Over 80% of data are fully complete, that's good. And there is much more missing ice_on then ice_off data, for some reason. There are no records with both missing, that's good, such records would be useless.

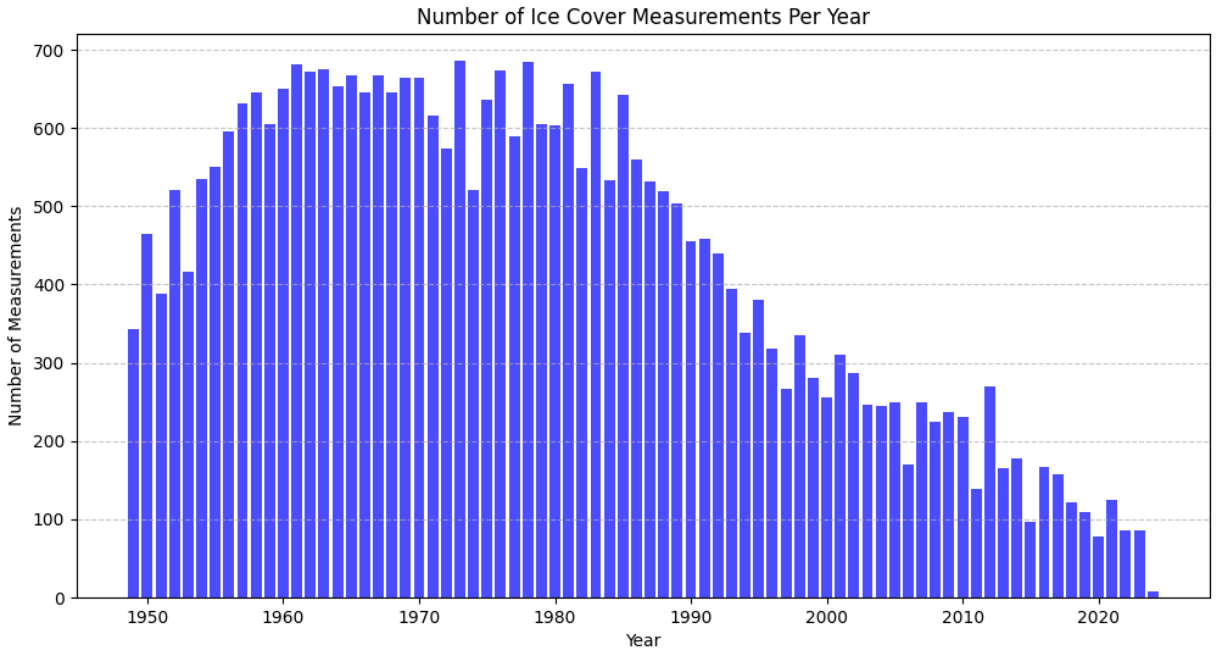
Next thing I wanted to look at, are the differences in the time that individual stations measured. It would be best if all stations measured from 1950 to 2024, but that's sadly not the case. Those time series are somewhat incomplete. I have plated two graphs, where we can see for each station the time range from when to when the individual stations collected data. They are ordered based on the maximal time in the first plot and based on the minimal time in the second.


Time Coverage of Lake Ice Data by Station
Stations (Ordered by First Year of Data)
Year
0
200
400
600
800
1000
1200
1950
1960
1970
1980
1990
2000
2010
2020

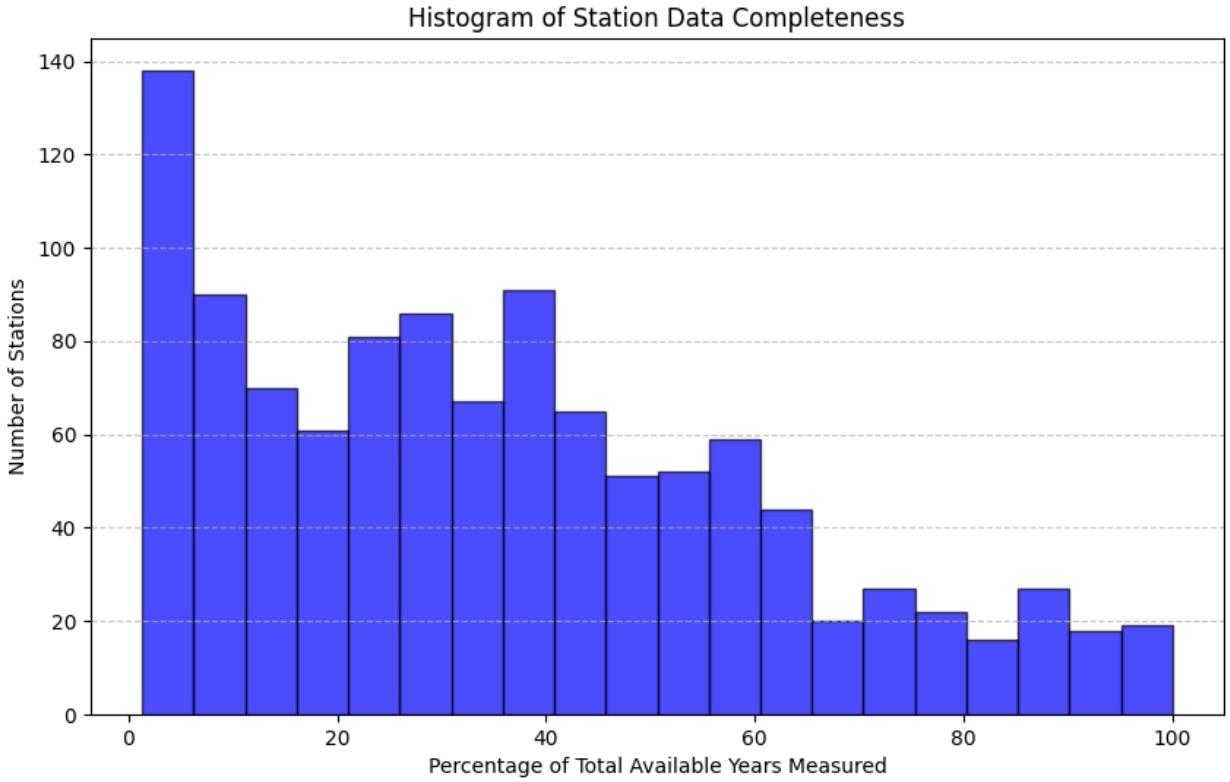


What concerns me a bit from this plot is that it seems that we don't have that much data from recent years which are for us the most interesting as we want to mainly study the effect of global warming in the recent years.

I have created two more plots investigating this time series incomplete. This following graph is a bar chart, where we can see how many measurements we have in given year. And the other graph investigating the length of those time series. If the whole time interval is from 1950 to 2024, then the histograms shows the distribution of the length of individual series from measurement stations in percent of this whole interval.

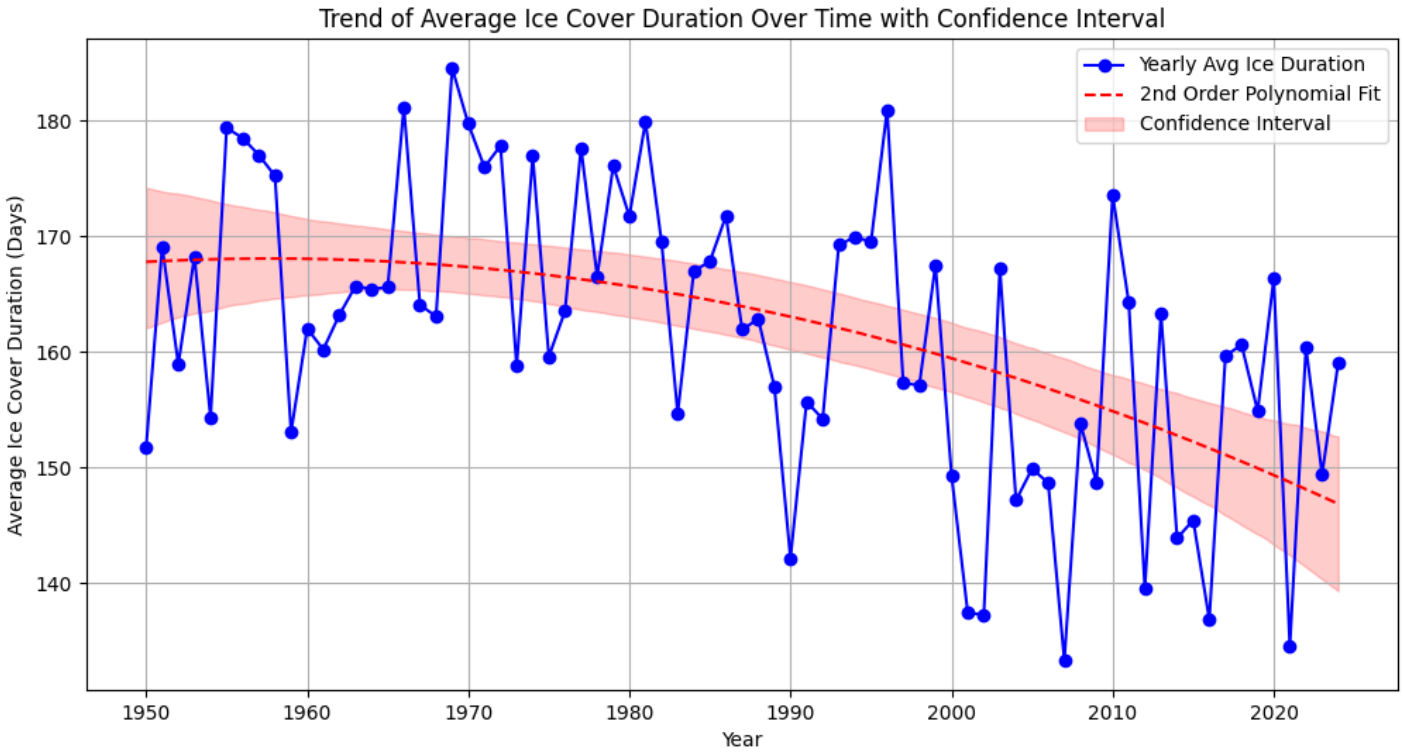


We can see here again, that we have much more old data then new data.

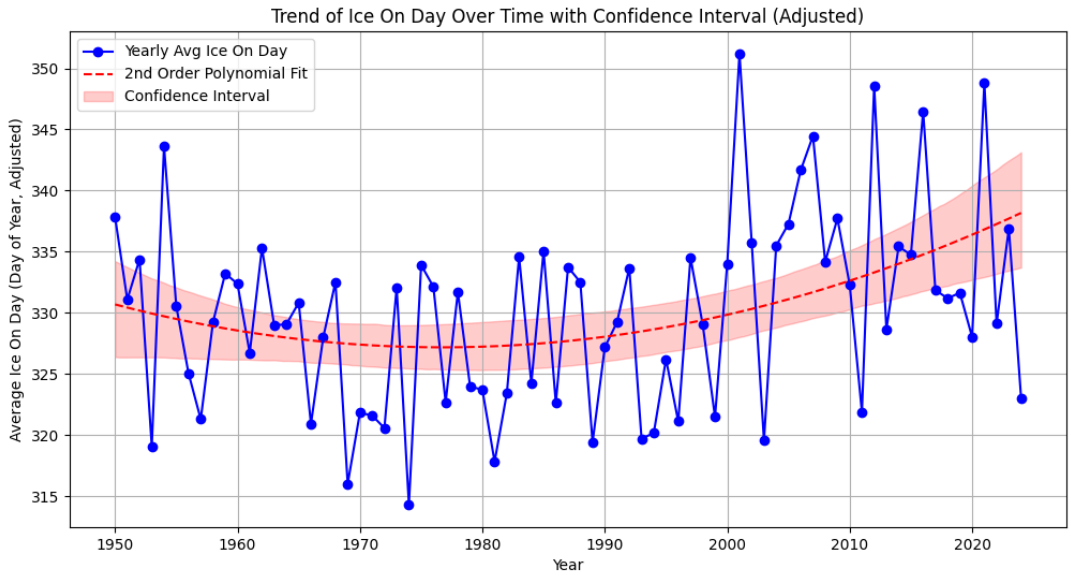


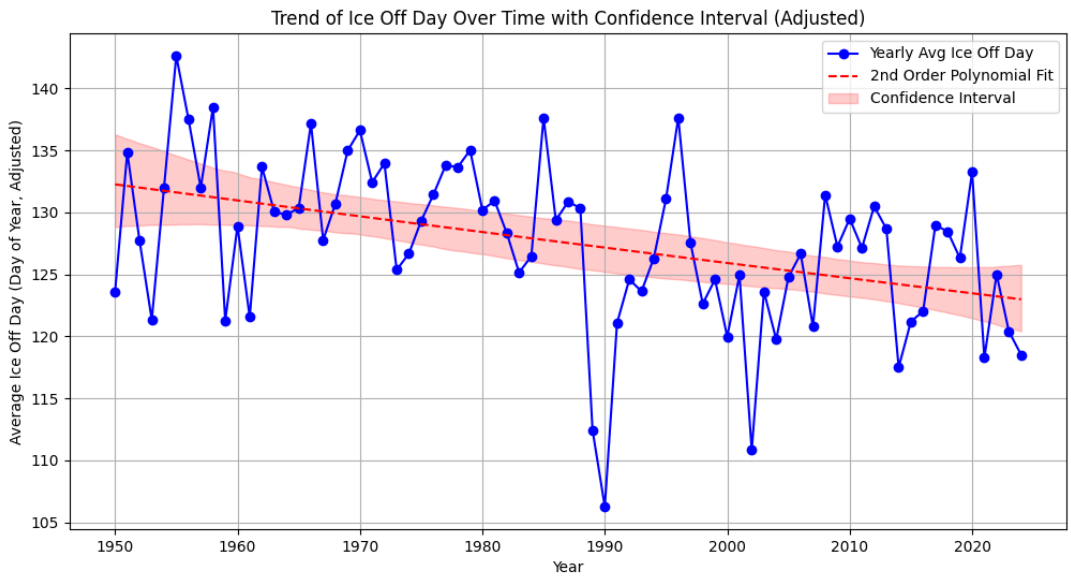
The amount of time series with given length decreases with the length, we have quite a lot very short time series, and quite a few very long like >65%.

Then I tried to do some simplest model possible to investigate the time dependence of the data. I have taken only the data without any missing values having both ice_on and ice_off and calculate the time difference ice_off - ice_on, which is like the length in days how long the lakes were frozen, and for each year, I have calculated just simple average of the ice cover duration in that year and I have fitted a second order polynomial. Some 95% interval was calculated using bootstrap. The result is quite nice, we can see quite well the acceleration of lake ice cover decline in recent years, but of course it's very naive model.



And I have done the same thing with only the ice_on and ice_off days. Bellow are the plots:





As expected the ice on day tends in increase in recent past and ice off decrease. The ice_on day dependence looks quadratic, whereas the ice_off function only linear.